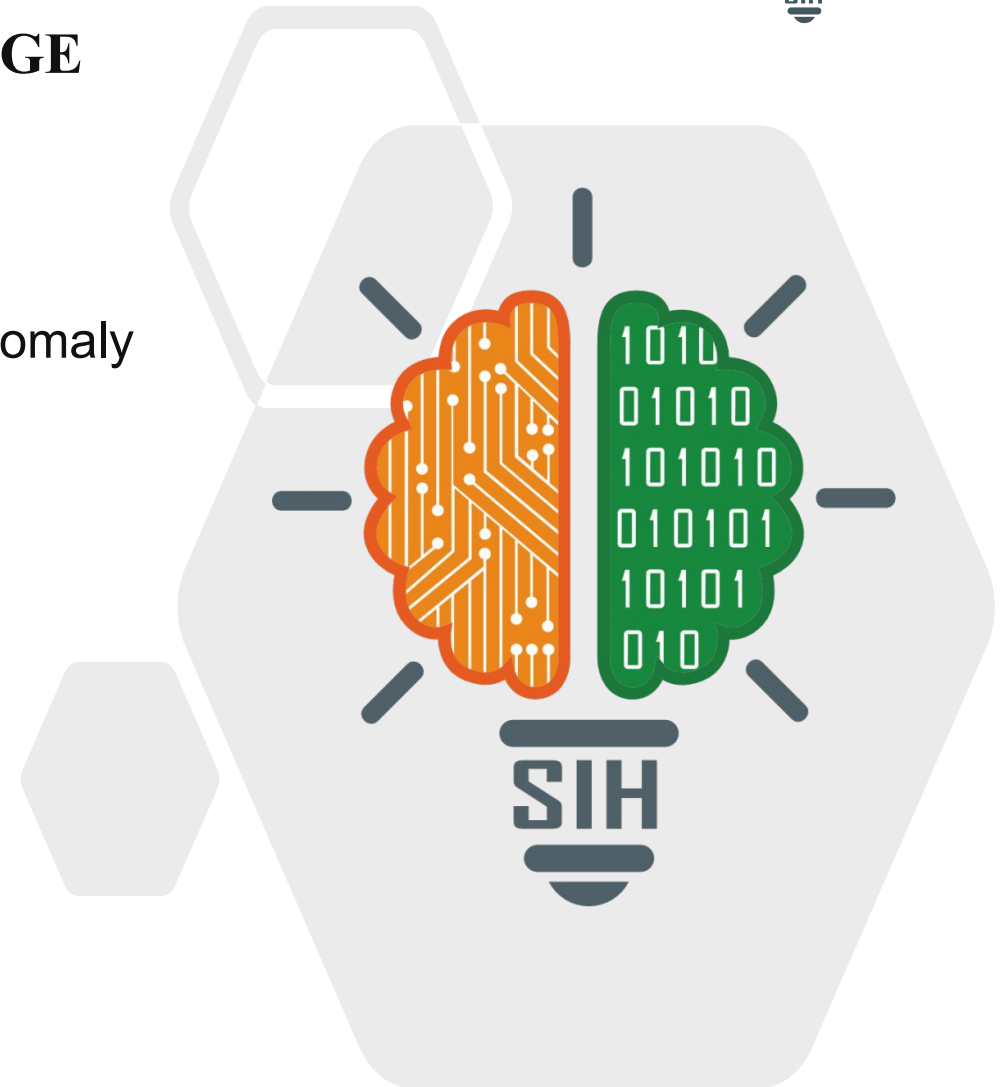


TITLE PAGE

- **Problem Statement ID** – SIH26073
- **Problem Statement Title** – AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)
- **Theme** – Disaster Management
- **PS Category** – Software
- **Team ID** – DM-SW-21
- **Team Name** – MeghDrishti



• Proposed Solution :-

A software-only weather-station quality platform that separates genuine extremes from sensor and telemetry faults.

• **Unified observation layer:-** Pulls station data and weather context into one auditable pipeline.

• **Hybrid anomaly detection:-** Uses physical rules, persistence, drift, spikes, dropout checks, and Isolation Forest.

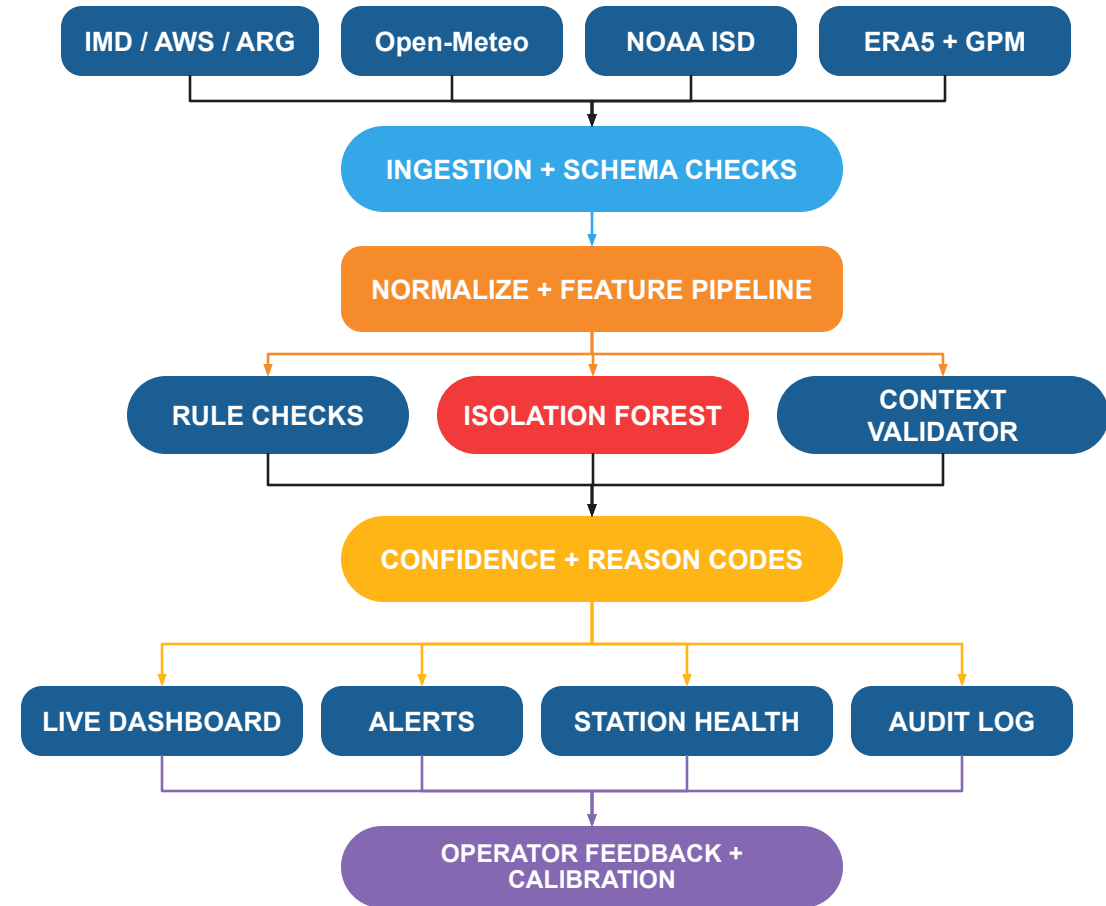
• **Extreme-vs-fault validation:-** Compares nearby stations, forecasts, ERA5, and GPM rainfall context.

• **Explainable review:-** Shows severity, confidence, evidence, and reason codes for each flag.

• **Safe data handling:-** Preserves raw observations and stores quality flags separately.

• **Operator learning loop:-** Confirmed labels improve thresholds and model calibration.

MeghDrishti – Weather Station Quality Intelligence



• Technologies to be Used:-

• **Data Sources:-** IMD API/Data Service, Open-Meteo, NOAA ISD, ERA5, NASA GPM IMERG

• **Backend:-** Python, FastAPI

• **Data Preparation:-** Pandas, NumPy

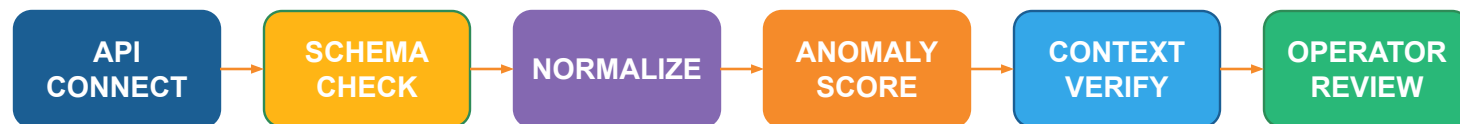
• **AI/ML:-** Rule checks, Isolation Forest, contextual scoring

• **Storage:-** PostgreSQL, Redis

• **Orchestration:-** Apache Airflow

• **Cloud/Infra:-** Docker, GitHub Actions, Grafana

• Process Flow :-



Fullforms:-

DATA SOURCES

IMD

OPEN-METEO

NOAA ISD

ERA5

NASA GPM

BACKEND / ML



Python



FastAPI



Isolation Forest



Pandas + NumPy

DATA LAYER



PostgreSQL



Redis



Apache Airflow

CLOUD / OPS



Docker



GitHub Actions



Grafana

Feasibility :-

- Technical: Python, Fast API, Airflow, PostgreSQL, and ML stack
- Economic: software-only deployment with open-source components
- Operational: 5–15 minute polling with retry and cached replay
- Data: official IMD path with independent context sources

Potential Challenges:-

- IMD access may require credentials or onboarding
- Historical sensor faults are rarely labelled
- Local extremes can resemble faulty readings
- Remote stations may lack spatial neighbors

Mitigation Strategies:-

- Adapter-based ingestion and cached demonstration data
- Synthetic injection of stuck, drift, spike, and dropout faults
- Rule, model, nearby-station, and weather-context voting
- Raw-data preservation with an operator audit trail

Viability:-

- Lightweight models can run continuously on modest cloud resources
- The architecture supports multiple station networks and vendors

Use Cases:-

- Real-time quality monitoring for IMD/AWS/ARG feeds
- Stuck sensor, drift, spike, dropout, and telemetry detection
- Station-health ranking for preventive maintenance
- Cleaner input data for forecasts and disaster review

Challenges:-

- API limits, latency, and intermittent connectivity
- Seasonal and regional concept drift
- False positives during genuine extreme weather
- Operator trust in machine-generated flags

Solutions:-

- Retries, caching, health checks, and replay queues
- Region and season-aware thresholds and calibration
- Context checks using nearby stations, forecasts, ERA5, and GPM
- Confidence scores, reason codes, and human confirmation

SUPPORTING FACTS FOR FEASIBILITY AND VIABILITY

- WMO guidance treats observation quality control as an operational requirement.
- ADT demonstrates near-real-time checks across mixed weather-station networks.
- NOAA ISD, ERA5, and NASA GPM provide independent validation context.

• Potential impact on the target audience:-

- IMD / operators: faster review of questionable stations
- Disaster teams: cleaner evidence during extreme weather
- Researchers: preserved raw data with reproducible QC flags
- Maintenance teams: ranked station-health priorities

• Key Operational Insights

- Unusual values require weather context before rejection.
- Reason codes make every flag reviewable.
- Human confirmation creates labels for later calibration.
- Raw observations stay available for audit and research.

• Benefits of the Solution (Operational, Scientific, Social, Economic)

Type	Benefit	Supporting Example
Operational	Faster station triage	Severity + reason codes
Scientific	Cleaner observation history	Raw values kept for audit
Social & safety	Stronger extreme-weather evidence	Context check before action
Economic	Lower manual review burden	Ranked maintenance priorities
Environmental	Fewer unnecessary site visits	Remote diagnosis first

• Unique Outcomes from Our Solution

- **Extreme-event protection**:- Context validation reduces the risk of rejecting genuine weather extremes.
- **Explainable anomaly evidence**:- Operators see confidence, reason codes, and the signals behind each alert.
- **Safer quality workflow**:- Flags never overwrite the original observation.
- **Continuous calibration**:- Confirmed reviews improve thresholds and model behavior.

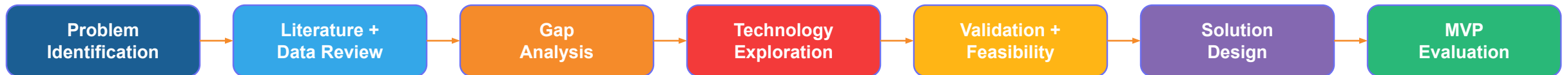
• Research and Data Sources:

- WMO observing systems and QA guidance - public.wmo.int
- ADT: Automatic Weather Station Data Tool - Frontiers in Climate
- NOAA Integrated Surface Database - ncei.noaa.gov
- Copernicus ERA5 reanalysis - cds.climate.copernicus.eu
- NASA GPM IMERG - gpm.nasa.gov/data

• Weather Data Platforms:

- IMD API Management Platform: api.imd.gov.in
- IMD Data Service Portal: dsp.imdpune.gov.in
- Open-Meteo API:** open-meteo.com

• Research Flow:



Feature	MeghDrishti	Rule-only QC	ML-only	Manual
1. Physical rule checks	✓	✓	✗	✓
2. Unsupervised anomaly model	✓	✗	✓	✗
3. Weather-context validation	✓	✗	✗	✓
4. Confidence and reason codes	✓	✓	✗	✗
5. Raw-data preservation	✓	✓	✓	✓
6. Human feedback loop	✓	✗	✗	✓
7. Continuous automated pipeline	✓	✓	✓	✗